



The Likelihood Function Transcript

Hello everyone, welcome to lesson 1 video 2. In this lesson, we will define the likelihood function, the central concept in maximum likelihood estimation. In this video, let's learn about how do we find out the value of the parameter, what mathematical technique do we use to find out the value of the parameter. So how do you use calculus and so on.

And so when the advantages, mathematical advantages that we have got to derive the value of the parameter is what we are going to look into this video. So the very first thing is you already know the data. What you have to do is you will have to create the likelihood function.

Once you create the likelihood function, which you say that it maximises the chances of getting the data, right. So let's say for example, a function which relates the value of the parameter and the data that you have got. So the value of the parameter given the data, this is what you will have to first create or write down.

From here, once we have that function, okay, once we have that likelihood function and we'll look into one of the examples. Once we have the functions, what we'll have to do is you already know to find out the maximum value of function, you will have to first take its derivative of the slope equals to zero because that is the place or at minima or wherever the change point is happening, right. So those are the places wherein the slope of the function is equals to zero.

So the very first thing that you'll do is to write down the likelihood function. Second is to take the derivative. Once you have taken the derivative, set it equals to zero, find out the points where the function's derivative equals to zero, take the second derivative test and find out whether it's a maxima or minima.

Okay, once you have found that it's a maximum point, you have actually got your maximum likelihood estimation. That's the entire method. Okay, but we generally do not maximise the likelihood function directly.

Instead, what we do is we take log likelihood or we maximise the log likelihood because there are product of quantities. Now, to take log, it becomes much more convenient to do the maths or mathematics becomes much more easier when we take natural log. That's why we generally go ahead and take log likelihood.

So maximum likelihood estimations are also called as log likelihood estimations. So why do we use that log function? The log function is monotonic and you already know that it is monotonic and which means that maximising $L(\theta)$ exactly is equals to maximising the log of $L(\theta)$. Okay, and because it has a lot of mathematical



convenience, hence we instead of maximising the likelihood function, we maximise log of likelihood function.

In summary, the likelihood function captures how probable the observed data are for different parameter values. It is a foundation of maximum likelihood estimation. In the next video, we'll see how to actually find MLE.

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